

What is Fourth Normal Form (4NF)?

Definition

A relation is in Fourth Normal Form (4NF) if:

- It is already in BCNF.
- It contains no non-trivial Multivalued Dependencies (MVDs).

For every Multivalued Dependency:

$$X \twoheadrightarrow Y$$

- X must be a Super Key.
- A table should not store multiple independent facts about the same entity.

Multivalued Dependency Review

Suppose an ECE student like Aadya has:

Skills:

- C++
- MATLAB

Languages:

- English
- Hindi

These two facts are completely independent.

Table

Student	Skill	Language
Aadya	C++	English
Aadya	C++	Hindi
Aadya	MATLAB	English
Aadya	MATLAB	Hindi

Notice what happened.
Every skill combines with every language.
This redundancy comes from two independent multivalued facts.

Decompose into

Original: | Student | Skill | Language |

StudentSkill

Student	Skill
Aadya	C++
Aadya	MATLAB

StudentLanguage

Student	Language
Aadya	English
Aadya	Hindi

No unnecessary combinations remain.
The relation is now in 4NF.

What is Fifth Normal Form?

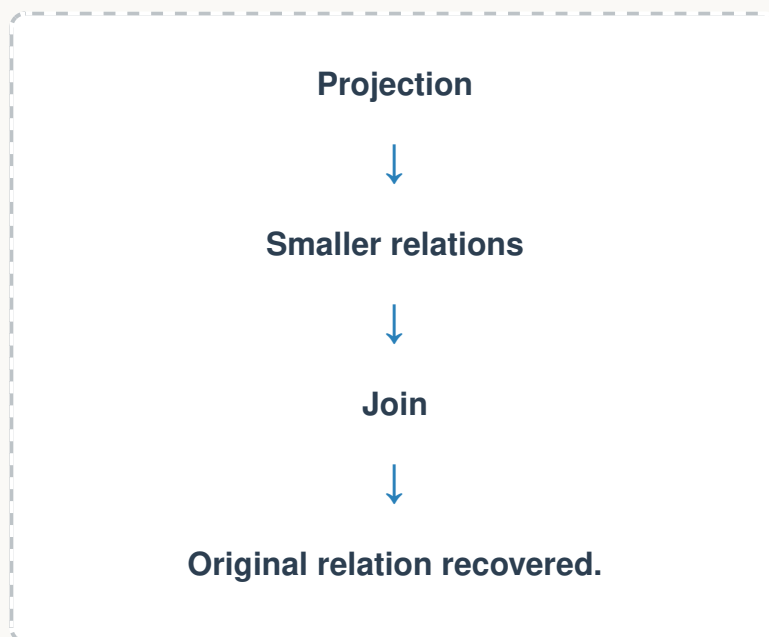
Definition

A relation is in Fifth Normal Form (5NF), also called Projection-Join Normal Form (PJNF), if:

- It is already in 4NF.
- Every non-trivial Join Dependency is implied by Candidate Keys.

Projection Join Normal Form (PJNF)

5NF is also called PJNF because:



Mathematical Condition for Lossless Join

Suppose:

$$R \rightarrow R_1 + R_2$$

The decomposition is lossless if:

$$(R_1 \cap R_2) \rightarrow R_1$$

OR

$$(R_1 \cap R_2) \rightarrow R_2$$

In words:

The common attributes between the decomposed relations must functionally determine at least one of the decomposed relations.